

THE CHARACTERISTIC DECAY WITH TIME OF SURFACE CHARGES ON DIELECTRICS

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Summary

With appropriate apparatus, the decay with time of electrical charge on the surface of Pyrex-glass, Perspex (PMMA) and Teflon (PTFE) was measured. To produce the surface charges, two different charging techniques were applied. One method utilised electrodes attached to the surface of the sample, the other consisted of the use of an ionisator (corona charging). Theoretical interpretation of the experimental results gives conductivities, or mobilities, and charge carrier concentrations, depending on the technique applied. The decay of surface charge is not exponential and so the half-life of the decay is not readily related to the relaxation time constant ϵ/σ . Herein is $\epsilon = \epsilon_0 \epsilon_r$ the total dielectric constant and σ the conductivity.

1. Introduction

Electrostatic charges play an important, but generally an undesired role in the production, working up, and storage of insulating materials, especially plastics. These charges often cause unexpected and, sometimes, dangerous effects. It is, therefore, of great importance to gain more insight into the generation of these charges and their dissipation into the material.

One method to study the breakdown mechanism is to measure conductivities and the storage of charge in the bulk of the material at high field strengths [1]. Determining conductivities of plastics in the classical way is a great problem, as the currents to be measured are very weak and the current becomes stationary after several hours.

2. Experimental

Surface charges on flat dielectrics were measured with an electrostatic probe (Fig. 1), which functions here as a potential divider.

This probe, (Keithly 2503), is over a coaxial electrometer cable connected

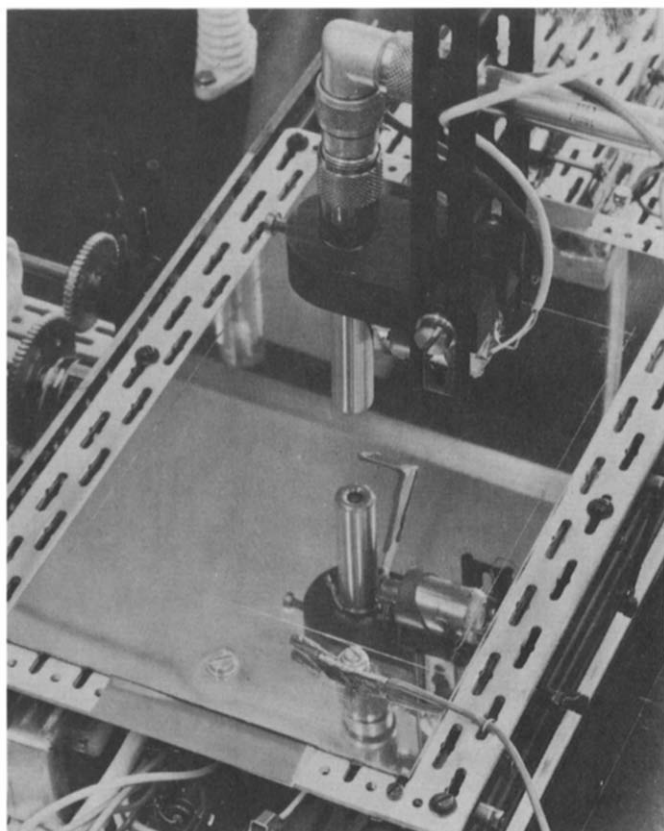


Fig. 1. The electrostatic probe.

to an electrometer, (Keithly 610 C), which drives the y -deflection of an x - y recorder. The flat dielectric ($12 \times 15 \text{ cm}^2$) is fastened on a mobile table, which is guided under the probe. The probe does not touch the dielectric. The table drives the x -deflection of the recorder. The dielectric can be positioned under a high voltage bridge, where it is charged with the ionisator (Fig. 2a) or H.V.-electrodes (Fig. 2b).

While charging with the H.V.-electrodes, the potential of these electrodes is raised in a very short time from earth potential to the high voltage level (5–15 kV). The discharging is started by earthing the electrodes. Figures 3a and 3b give an example of a series of characteristics for the charging and discharging, respectively, of a perspex dielectric. In these characteristics, the measured dielectric potential is given as a function of the x co-ordinate and with the time t as a parameter. The voltage measured is a measure of the surface charge density. For this purpose, the probe is first gauged in a special way. Measurement using the probe has been described more extensively in a previous paper [2].



Fig. 2. (a) Charging with an ionisator. (b) Charging with electrodes.

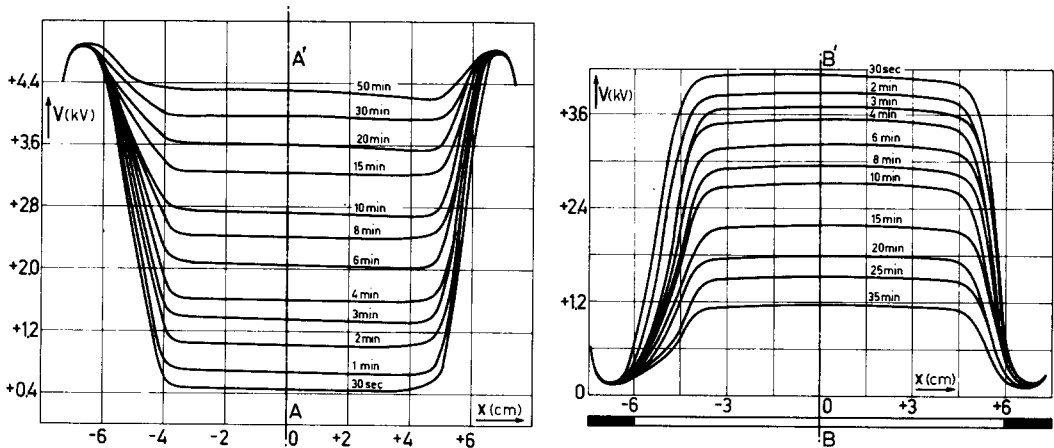


Fig. 3. (a) Charging-characteristic of Perspex with electrodes, thickness 1.1 mm; distance probe to dielectric is 0.5 cm. (b) Discharging-characteristic of Perspex with electrodes, thickness 1.1 mm; distance probe to dielectric is 0.5 cm.

The charging with the ionisator is done in an analog way. The dielectric is charged nearly homogeneously to the desired level (5–15 kV). Examples of decay characteristics for Pyrex-glass, Perspex and Teflon are given in Figs. 4, 5 and 6. It is notable that, with the exception of the glass, the dielectrics show characteristics which become distorted only a short time after the decay has commenced. The earth side of each dielectric was covered with a silver-painted electrode having an area considerably smaller than the total area of the dielectric.

3. Theory

Firstly, it will be shown that, in the given measurements, the decay of charge is caused by the conduction in the bulk, or over the surface of the material. All experiments were carried out in the open air. Neutralisation by free-air ions is of no influence for two reasons. The first is that this ion concentration is very small. The second is that the electric field strength at the surface of the dielectric is very small. A theoretical calculation shows that the electric field

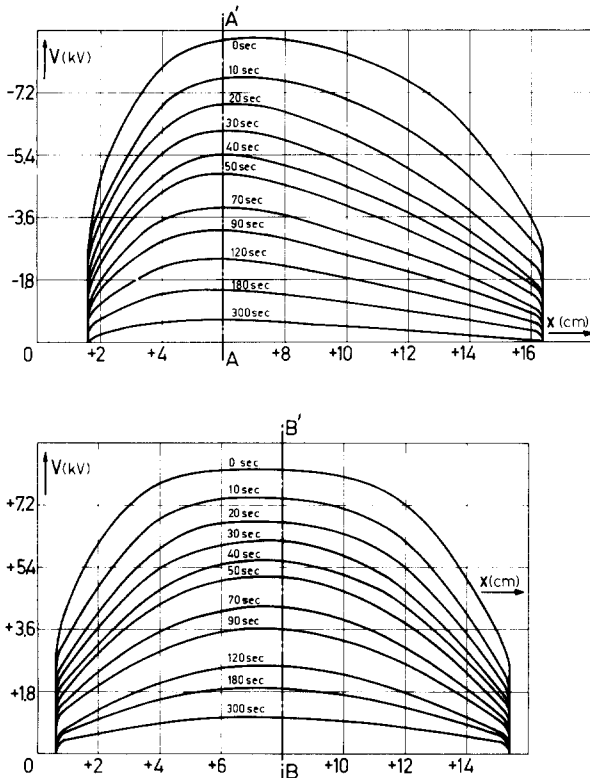


Fig. 4. Decay-characteristics of Pyrex-glass, corona-charging, thickness 10.2 mm; distance probe to dielectric is 2 cm.

strength at the surface of a perspex-dielectric, thickness 3.2 mm, is 3 % of its value inside the material [2]. Measurements have shown that, in the case of a perspex-dielectric isolated from earth, a unipolar charge decays 1/3 of its initial charge in 20 hours [3]. A calculation with the results given above shows that, in our case, the half-life of decay, caused by the neutralisation, would take thousands of hours. Another influencing factor may be the polarization of the dielectric. One can show, mathematically, that this influence is negligible, as long as it is valid that every relaxation time constant is much smaller than the half-life of decay. This seemed to be valid for the dielectrics used in our experiments: Pyrex-glass, Perspex and Teflon.

3.1. Electrode-charging

Now we will make a theoretical model of the cross-section of the experimental set-up given in Fig. 2b. Here, it is assumed that the model can be one-dimensional with the independent variable x . This is realistic if the dimensions are suitably chosen. Further, it is assumed that $\rho_T \gg \rho_P$ where ρ_T is the specific resistance of the Teflon insulation and ρ_P is the specific resistance of Perspex. For a segment of thickness dx , an equivalent electrical circuit can be given ac-

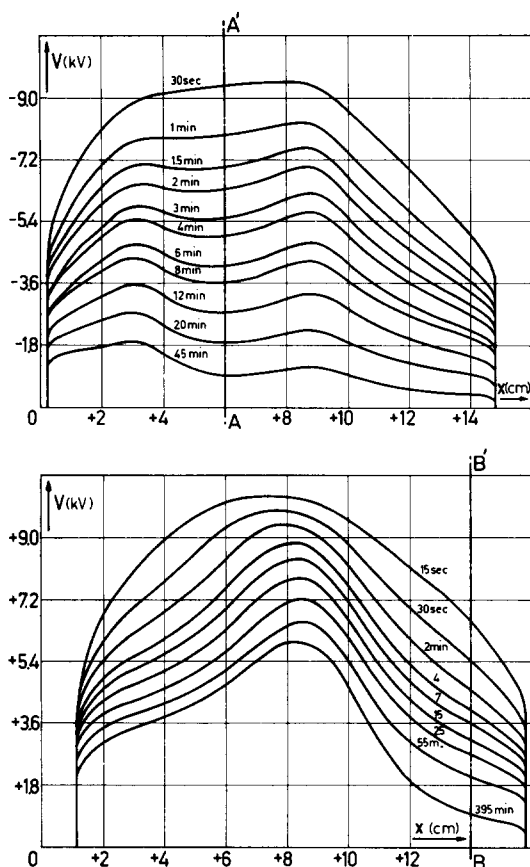


Fig. 5. Decay-characteristics of Perspex, corona-charging, thickness 1.1 mm; distance probe to dielectric is 2 cm.

cording to Fig. 7. For the node K the sum of the currents I has to be zero: $\Sigma I = 0$. Neglecting magnetic effects, one obtains the following differential equation:

$$\frac{\partial V}{\partial t} = a \frac{\partial^2 V}{\partial x^2} - \frac{V}{\tau} \quad (1)$$

$$\text{with } a = \frac{h}{\epsilon} \sigma_p \quad \text{and} \quad \tau = \epsilon \rho_T$$

and where σ_p is the specific surface conductivity of Perspex, ϵ is the total dielectric constant of Teflon, and h is the thickness of the Teflon insulation.

The differential equation (1) is among others known in the theory of heat transport [4].

In the case of the charging of the perspex-dielectric with the initial condition $v(x, 0) = 0$ and the boundary conditions $v(-l, t) = V$ and $v(+l, t) = V$, one finds the solution:

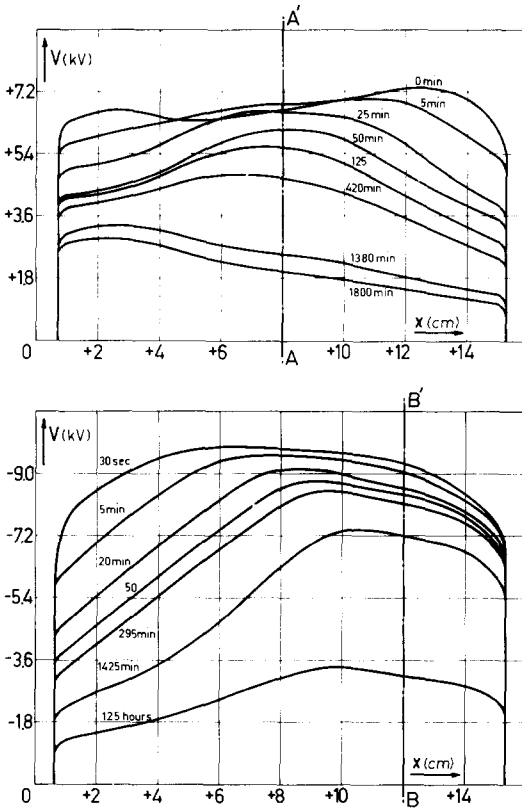


Fig. 6. Decay-characteristics of Teflon, corona-charging, thickness 1 mm; distance probe to dielectric is 2 cm.

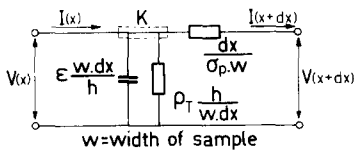


Fig. 7. Equivalent circuit for a segment of the dielectric.

$$v = V \frac{\cosh bx}{\cosh bl} \left\{ 1 - \exp(-t/\tau) \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \exp[-a(2n+1)^2 \pi^2 t/4l^2] \cos \frac{(2n+1)\pi x}{2l} \right\} \quad (2)$$

with $b = 1/\sqrt{\sigma_p \rho_T h}$.

In the case of the discharging of the perspex-dielectric with the initial condition the stationary state after charging:

$$v(x,0) = V \frac{\cosh bx}{\cosh bl}$$

and the boundary conditions $v(-l, t) = 0$ and $v(+l, t) = 0$ it follows

$$v = V \frac{\cosh bx}{\cosh bl} \exp(-t/\tau) \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \exp[-a(2n+1)^2 \pi^2 t/4l^2] \cos \frac{(2n+1)\pi x}{2l} \quad (3)$$

In the case that the conduction by the Teflon insulator is much smaller than that through the Perspex insulator, which is valid if $\rho_{Th} \gg \sigma_p$ and $\tau \rightarrow \infty$ the solutions (2) and (3) simplify, respectively, to the solutions (4) and (5), given below.

$$v = V \left\{ 1 - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \exp[-a(2n+1)^2 \pi^2 t/4l^2] \cos \frac{(2n+1)\pi x}{2l} \right\} \quad (4)$$

$$v = V \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \exp[-a(2n+1)^2 \pi^2 t/4l^2] \cos \frac{(2n+1)\pi x}{2l} \quad (5)$$

3.2. Corona-charging

In the next case, we will consider models which are applicable to the flat dielectric after corona-charging. For this, we consider the situation according to Fig. 8. It shows an ideal insulator which is charged homogeneously at the starting point $t = 0$. There are no thermal free carriers ($n = 0$). Then it follows for the surface potential v with the aid of the law of conservation of charge and the law of Poisson [5] for the two time intervals I and II:

$$\text{I. } 0 < t < \frac{d^2}{V_0 \mu} \quad v = V_0 \left(1 - \frac{V_0 \mu t}{2d^2} \right) \quad (6)$$

$$\text{II. } t > \frac{d^2}{V_0 \mu} \quad v = d^2/2\mu t$$

where μ is the mobility of the charge carriers and $V_0 = v(t = 0)$.

In the time interval I charge is injected from the surface into the bulk of the material and a charge-front moves towards the earthed plane. In the time interval II, the surface charge decays by a space charge limited current (SCLC) [6].

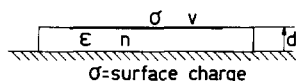


Fig. 8. Flat dielectric on an earthed plane.

In the case when the material has thermal free carriers, then an approximate solution for the surface potential v is [7]:

$$V_0/v = \frac{1 + \epsilon V_0}{2ned^2} \exp(-t/\tau) - \frac{\epsilon V_0}{2ned^2} \quad (7)$$

with $\tau = \epsilon/ne\mu$, and where e is the elementary charge, 1.6×10^{-19} C. In deriving eqn. (7), the contribution of the space charge to the surface potential has been neglected. Kanazawa *et al.* [8] gave a correct mathematical solution, which is, however, not as practical as solution (7).

It now seems that, often, the time interval I is passed when the corona-charging is stopped. In this case, it is more practical, for the solution in the time interval II, to do the time transformation: $t' = t - d^2/V_0\mu$ and to express the inverse value of the potential v . It follows that:

$$\frac{1}{2} V_0/v = 1 + \frac{\mu V_0}{d^2} t \quad (8)$$

The charge storage during the corona-charging can be calculated from the continuity between charging and discharging. Therefore, measurements of the charging currents and decay characteristics were made with some polymers by Takada *et al.* [1].

4. Experimental results

4.1. Electrode charging

Comparing the path of the curves of Fig. 3 with those of the solutions (4) and (5) (see ref. [4]), one can state that the curves of Fig. 3 are much more flatter. This is the consequence of the integrating function of the probe, that is, the contribution to the probe potential is not only provided by the charges on the spot but also by those in the surroundings.

The contribution to the probe potential diminishes as the charges are more distant from the probe. There is a limited distance for the charges to contrib-

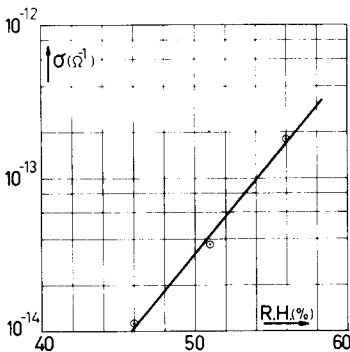


Fig. 9. Specific surface conductivity as a function of relative humidity.

ute. The solutions (4) and (5) are applicable if one compares the voltages in the centre ($x = 0$, in Fig. 3 the cross-sections AA' and BB'). Once the constant a is determined, the specific surface conductivity σ_p can be calculated. In the case of Perspex, the conduction along the surface is much greater than in the bulk, because there is an important moisture absorption. The moisture does not easily penetrate into the bulk. The average values of σ_p calculated from the charge and discharge characteristic for positive, as well as for negative voltage, are plotted in Fig. 9 as a function of the relative humidity. The curve shows an exponential relationship between relative humidity and specific surface conductivity. The same result is given by Sutton [9] for the relationship between the surface conductivity of glass and relative humidity. Shashoua [10] found this result for the relationship between the half-lives of decay of several polymers (polyvinylfluoride, Nylon 6-6, etc.) and relative humidity.

4.2. Corona-charging

The voltages on the cross-sections AA' and BB' of Figs. 4, 5 and 6 are plotted in Figs. 10, 11 and 12 as a function of time for Pyrex-glass (inverse values), Perspex (inverse values), and Teflon (linear), respectively. Applying eqns. (7),

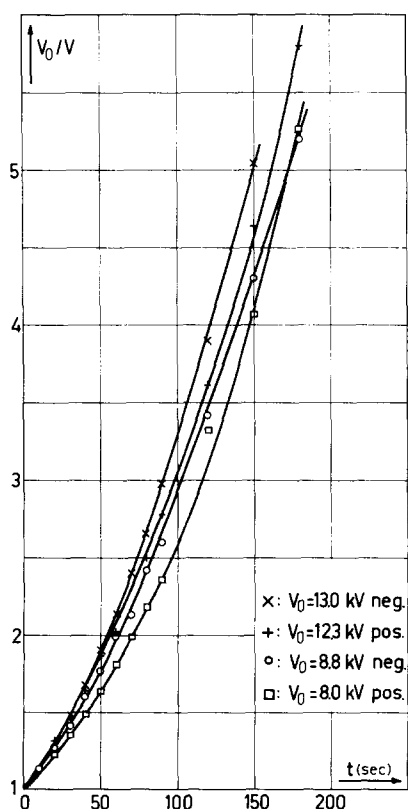


Fig. 10. Decay of surface potential in time for Pyrex-glass (after Fig. 4).

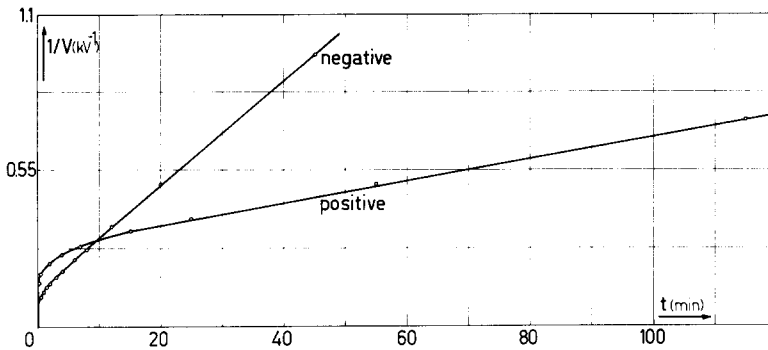


Fig. 11. Decay of surface potential in time for Perspex (after Fig. 5).

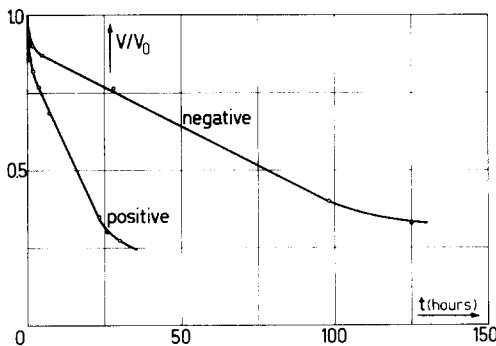


Fig. 12. Decay of surface potential in time for Teflon (after Fig. 6).

(8) and (6) for Pyrex-glass, Perspex, and Teflon, respectively, we found for each dielectric a suitable theoretical model. The results are summarized in Table 1. In the case of Pyrex-glass the presence of thermal free carriers is assumed. The decay is then faster as compared with the case where free carriers are not present (the characteristics of Fig. 10 would then show a straight line). It has to be mentioned, however, that, in principle, the same effect could be established by surface conduction. When both sides of the dielectric are covered with silver-painted electrodes and the dielectric is charged by connecting these electrodes with a supply voltage, there is no marked change in the decay compared with the decay in the case of corona-charging as described above.

5. Discussion

As our experiments have shown, the known theories concerning the decay of the charge on dielectrics are applicable. All these theories are based on the existence of only one type of charge carrier, and this is a restriction. For it is known that some dielectrics have several types of charge carriers, mostly

TABLE 1

Results of measurements (corona-charging method)

Dielectric	Mobility $\left(\frac{\text{m}^2}{\text{V sec}}\right)$	Character of conduction		Half-life of decay		ρ (Ωm)	ϵ_r
		SCLC	thermal free carriers	average measured	from the relaxation time constant, $\epsilon\rho\ln 2$		
Pyrex-glass	2×10^{-10} — 2×10^{-11}	immediately	yes	1.5 min	1.7 min	2.1×10^{12}	8.2
Perspex (PMMA)	4×10^{-12} — 8×10^{-14}	immediately	no	15 min	2.6 hours	4.1×10^{14}	3.7
Teflon (PTFE)	0.4 — 2.6×10^{-15}	later	no	50 hours	7 hours	1.7×10^{15}	2.1

Relative humidity 52 %—58 %, temperature 21°C, open air

caused by impurities. The mobility, as well as the charge carrier concentration, can be determined by the given techniques and this is the great advantage over, for example, conductivity measurements, which give, in most cases, only a value for the product of mobility and carrier concentration. This has already been indicated by Davies [11]. Determining the type of charge carrier is sometimes possible with other techniques (for example, the experiments with photo-emission), but this is not a rule [12]. The large spread in the determined mobilities is generally found, and, at the moment, there is no satisfactory explanation. There are a number of parameters influencing the decay. It has been shown that, during production and working up, considerable charge can be stored inside polymers, while additives can determine the charge stored, as well as the conductivities, to a large degree [13]. Conditioning of the samples will be of great importance (storage time, relative humidity, temperature, contamination). Some parameters have been ruled out as not playing an important role under a certain condition, as, for example, diffusion [14]. Of great interest are the results obtained with pulse methods. Here, mobilities are found which are considerably higher, as in the decay experiments [15]. Davies [7, 11] charged dielectrics by contacting them with metals in vacuum. His decay characteristics and the decay characteristics of our corona-charging are applicable on the same theoretical model. This shows that, in this case, the charging technique is not essential for the decay.

Table 1 also presents the half-lives of decay which can be calculated from an assumed exponential decay with the relaxation time constant $\epsilon\rho$. The half-life of decay is now $\epsilon\rho\ln 2$. The relative dielectric constant ϵ_r and the specific bulk resistance ρ were determined by separate measurements. The measurements agreed with the values known in literature. The real half-lives of decay are determined by taking the average value of some decay-characteristics (positive as well as negative) of each dielectric. Comparing the measured half-lives of decay with the calculated ones, it can be said that the relaxation time constant has no predicting significance in what concerns the duration of the decay.

6. Conclusions

(1) Neutralisation of surface charge by ions from the air is not of important significance if the duration of the experiments is smaller than 1000 hours, for field strengths up to 100 kV/cm. (In our experiments: a dielectric of 1 mm thickness charged up to 10 kV.)

(2) The surface resistance of Perspex is very sensitive to water and is exponentially related to the relative humidity. The dependence of the bulk resistance on the relative humidity seems to be much slighter.

(3) Teflon approaches the ideal insulator more than other commercial dielectrics because of its lack of thermal free charge carriers and the low mobility of the injected charge carriers.

(4) The decay of surface charge is not exponential and the relaxation time constant is of no practical significance for the real decay time.

(5) Decay characteristics are valuable for studying the conduction in dielectrics.

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